

Complete intersections of Hodge level one

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Introduction

Throughout this article, we shall denote by $n = 2m + 1$ a fixed odd integer ≥ 3 , and by $\underline{a} = (a_1, \dots, a_v)$ a fixed sequence of integers > 1 , such that the smooth complete intersections

$$V_n(\underline{a}) \subset \mathbb{P}^{n+v}(\mathbb{C})$$

of dimension n and multidegree $\underline{a}$ have Hodge level 1. This condition means that, for such a complete intersection X , the intersection of v hypersurfaces of degrees $a_1, \dots, a_v$ which, at the points of X , are smooth and meet transversely,

- a) for $p + q = n$ and $|q - p| > 1$, one has $H^q(X, \Omega_X^p) = 0$;
- b) $H^m(X, \Omega_X^{m+1}) \neq 0$.

Recall that, by Lefschetz,

- c) $H^*(X, \mathbb{Z})$ is torsion-free;
- d) for k odd, $k \neq n$, one has $H^k(X, \mathbb{Z}) = 0$;
- e) for $p \leq n$, $H^{2p}(X, \mathbb{Z})$ has rank 1 and is purely of type (p, p) ; if $\eta \in H^2(X, \mathbb{Z})$ is the cohomology class of a hyperplane section, then $H^{2p}(X, \mathbb{Q}) = \mathbb{Q} \cdot \eta^p$.

The list of possible values of $(n, \underline{a})$ appears in [10].

The *intermediate Jacobian* $J(X)$ of X is, by definition, the complex torus

$$J(X) = H^n(X, \mathbb{Z}) \backslash H^n(X, \mathbb{C}) / H^{m+1, m}(X) = H^{m, m+1} / H^n(X, \mathbb{Z}).$$

The cup product on $H^n(X, \mathbb{Z})$ defines a Riemann form on $J(X)$, which is therefore an abelian variety polarized in the principal series (=principally polarized). We show that the construction $X \mapsto J(X)$ is “purely algebraic”, i.e. independent of the topology of $\mathbb{C}$. In particular, if X is defined over a field $k \subset \mathbb{C}$, then $J(X)$ is defined over the same field k . In the case $n = 1$ (which we exclude below only for reasons of convenience), $J(X)$ is the Jacobian of X . For $V_3(3)$, $J(X)$ is the Jacobian of the Fano surface of X (see [1]). For $n = 3$ and X unirational, Manin’s methods [8] make it possible to express $J(X)$ using Jacobians. For $V_n(2, 2)$, if $(L_t)_{t \in \mathbb{P}^1}$ is

a Lefschetz pencil of hyperplane sections, and if C is the covering of $\mathbb{P}^1$ whose points are the linear subspaces of dimension $(n-1)/2$ in the L_t , then $J(X)$ is, up to isogeny, a direct factor of the Jacobian of C . In the other cases, I know no direct interpretation of $J(X)$. According to Lieberman [7], the Hodge conjecture implies that $J(X)$ is the quotient of the set of algebraic cycles on X , of codimension $(n+1)/2$ and homologous to zero, by a suitable equivalence relation.

As a corollary of the theory, one obtains (for n and $\underline{a}$ fixed as above) that smooth complete intersections of dimension n and multidegree $\underline{a}$ over a finite field $\mathbb{F}_q$ satisfy the Weil conjecture. For $V_5(3)$ and $V_3(4)$, this result is, to my knowledge, new.

1. Over $\mathbb{C}$

1.1. Let X be a nonsingular connected projective complex algebraic variety of dimension k . Duality theory for coherent algebraic sheaves provides an isomorphism (purely algebraic)

$$H^k(X, \Omega_X^k) \xrightarrow{\sim} \mathbb{C}. \quad (1.1.1)$$

On the other hand, Hodge theory (here GAGA + the Poincaré lemma) provides an isomorphism

$$H^k(X, \Omega_X^k) \xrightarrow{\sim} H^{2k}(X, \mathbb{C}). \quad (1.1.2)$$

The image under (1.1.1) of integral cohomology $H^{2k}(X, \mathbb{Z}) \subset H^{2k}(X, \mathbb{C})$ is the lattice $(2\pi i)^{-k} \mathbb{Z} \subset \mathbb{C}$.

This leads to the definition of the *Tate Hodge structure* $\mathbb{Z}(s)$ as the rank-one Hodge structure, pure of type $(-s, -s)$, with integral lattice

$$\mathbb{Z}(s)_{\mathbb{Z}} = (2\pi i)^s \mathbb{Z} \subset \mathbb{Z}(s)_{\mathbb{C}} = \mathbb{C}.$$

The cup product defines a morphism of Hodge structures

$$H^k(X, \mathbb{Z}) \otimes H^k(X, \mathbb{Z}) \longrightarrow \mathbb{Z}(-k). \quad (1.1.3)$$

The analogues of (1.1.3) in ℓ -adic or De Rham cohomology are as follows.

1.2. Recall that $\mathbb{Z}/r(1)$ denotes the group of r -th roots of unity. We identify $\mathbb{Z}(1)_{\mathbb{Z}/r} =_{\text{dfn}} \mathbb{Z}(1)_{\mathbb{Z}} \otimes \mathbb{Z}/r$ with $\mathbb{Z}/r(1)$ by the map $x \mapsto \exp(x/r)$. For every prime number ℓ , put

$$\mathbb{Z}_{\ell}(1) = \varprojlim_r \mathbb{Z}/\ell^r(1) \quad (\text{transition maps } x \mapsto x^{\ell^{r'-r}}).$$

The exponential map identifies $\mathbb{Z}(1)_{\mathbb{Z}_\ell}$ with $\mathbb{Z}_\ell(1)$ and $\mathbb{Z}(s)_{\mathbb{Z}_\ell}$ with

$$\mathbb{Z}_\ell(s) =_{\text{dfn}} \mathbb{Z}_\ell(1)^{\otimes s}.$$

If (1.1.3) is tensored with $\mathbb{Z}_\ell$, the resulting morphism

$$H^k(X, \mathbb{Z}_\ell) \otimes H^k(X, \mathbb{Z}_\ell) \longrightarrow \mathbb{Z}_\ell(-k) \quad (1.2.1)$$

is just the cup product in ℓ -adic cohomology. It has a purely algebraic meaning.

1.3. One has

$$H^k(X, \mathbb{Z}) \otimes \mathbb{C} = H^k(X, \Omega_X^*) =_{\text{dfn}} H_{\text{DR}}^k(X). \quad (1.3.1)$$

Here and below, we shall always identify $\mathbb{Z}(k)_{\mathbb{C}} = \mathbb{Z}(k)_{\mathbb{Z}} \otimes \mathbb{C}$ with $\mathbb{C}$ by the “defining” isomorphism that sends the integral lattice to $(2\pi i)^k \mathbb{Z}$. Tensor (1.1.3) with $\mathbb{C}$. The resulting morphism

$$H_{\text{DR}}^k(X) \otimes H_{\text{DR}}^k(X) \longrightarrow \mathbb{C} \quad (1.3.2)$$

is just the cup product in De Rham cohomology. It has a purely algebraic meaning.

1.4. Take X to be a complete intersection $V_n(\underline{a})$. Tensor (1.1.3) by $\mathbb{Z}(2m)$ and multiply the resulting morphism by $(-1)^m$. One obtains

$$\psi : H^n(X)(m) \otimes H^n(X)(m) \longrightarrow \mathbb{Z}(-1). \quad (1.4.1)$$

By hypothesis, the Hodge structure $H^n(X)(m)$ is of type $(0, 1) + (1, 0)$. The form ψ is a polarization of $H^n(X)(m)$, and, by Poincaré duality for X , the alternating form $\psi_{\mathbb{Z}}$ on the integral lattice $H^n(X)(m)_{\mathbb{Z}}$ has discriminant one.

There therefore exists a (unique) abelian variety polarized in the principal series $J(X)$, endowed with an isomorphism of polarized Hodge structures

$$\alpha : H^n(X)(m) \xrightarrow{\sim} H^1(J(X)).$$

We call $J(X)$ the *intermediate Jacobian* of X .

1.5. Let S be a smooth scheme of finite type over $\mathbb{C}$, and let $f : X \rightarrow S$ be a proper smooth morphism whose fibers are complete intersections $V_n(\underline{a})$. Then $R^n f_* \mathbb{Z}(m)$ is a variation of polarized Hodge structures on S , of type $(0, 1) + (1, 0)$ (see [5] I 2). By a theorem of Borel (see [4] 5.1 and 5.2 for the statement), it defines an abelian scheme polarized in the principal series $u : J(X/S) \rightarrow S$, endowed with an isomorphism

of variations of polarized Hodge structures

$$\alpha : R^n f_* \mathbb{Z}(m) \xrightarrow{\sim} R^1 u_* \mathbb{Z}.$$

This construction expresses that $J(X)$ varies algebraically with X ; it would not be difficult to extend it to the case where S is an arbitrary scheme over $\mathbb{C}$.

The analogues in ℓ -adic or De Rham cohomology of the variation of Hodge structures $R^n f_* \mathbb{Z}(m)$ and of α are as follows.

1.6. Tensoring with $\mathbb{Z}_\ell$, α defines an isomorphism of $\mathbb{Z}_\ell$ -sheaves

$$\alpha_\ell : R^n f_* \mathbb{Z}_\ell(m) \xrightarrow{\sim} R^1 u_* \mathbb{Z}_\ell. \quad (1.6.1)$$

The cup product in ℓ -adic cohomology, multiplied by $(-1)^m$, is a morphism of $\mathbb{Z}_\ell$ -sheaves

$$\psi_\ell : R^n f_* \mathbb{Z}_\ell(1) \otimes R^n f_* \mathbb{Z}_\ell(1) \longrightarrow \mathbb{Z}_\ell(-1) \quad (1.6.2)$$

and α_ℓ transforms ψ_ℓ into the polarization form

$${}_s\psi_\ell : R^1 u_* \mathbb{Z}_\ell \otimes R^1 u_* \mathbb{Z}_\ell \longrightarrow \mathbb{Z}_\ell(-1). \quad (1.6.3)$$

The forms (1.6.2) and (1.6.3) have a purely algebraic meaning.

1.7. The locally free coherent analytic sheaf $R^n f_* \mathbb{Z} \otimes \mathcal{O}$ underlies the coherent algebraic sheaf

$$R^n f_* \Omega_{X/S}^* =_{\text{dfn}} H_{\text{DR}}^n(X/S).$$

The evident integrable connection on $R^n f_* \mathbb{Z} \otimes \mathcal{O}$ is identified with the Gauss–Manin connection on $H_{\text{DR}}^n(X/S)$. Tensoring with $\mathcal{O}$, α defines an isomorphism of analytic sheaves

$$\alpha_{\text{DR}} : H_{\text{DR}}^n(X/S) \xrightarrow{\sim} H_{\text{DR}}^1(J(X/S)/S). \quad (1.7.1)$$

The Gauss–Manin connections of $H_{\text{DR}}^n(X/S)$ and $H_{\text{DR}}^1(J(X/S)/S)$ are regular ([5] I 4 or [2] II §7), and α_{DR} is a horizontal isomorphism. Hence α_{DR} is algebraic ([2] II 5.9).

The cup product in De Rham cohomology, multiplied by $(-1)^m$, is a horizontal morphism

$$\psi_{\text{DR}} : H_{\text{DR}}^n(X/S) \otimes H_{\text{DR}}^n(X/S) \longrightarrow \mathcal{O}_S \quad (1.7.2)$$

and α_{DR} transforms ψ_{DR} into the polarization form

$${}_s\psi_{\text{DR}} : H_{\text{DR}}^1(J(X/S)/S) \otimes H_{\text{DR}}^1(J(X/S)/S) \longrightarrow \mathcal{O}_S. \quad (1.7.3)$$

1.8. Let $S_{\mathbb{Z}}$ be the subscheme of the Hilbert scheme (over $\mathbb{Z}$) that parametrizes smooth complete intersections of dimension n and multidegree $\underline{a}$ in $\mathbb{P}^{n+v}$. Let $f : X_{\mathbb{Z}} \rightarrow S_{\mathbb{Z}}$ be the “universal” family of $V_n(\underline{a})$ parametrized by $S_{\mathbb{Z}}$. For every ring A , we shall denote by $f : X_A \rightarrow S_A$ the morphism obtained from f by extension of scalars from $\mathbb{Z}$ to A .

Put $G = \mathrm{PGL}_{n+v+1}$. The group scheme G (over $\mathbb{Z}$) acts on $X_{\mathbb{Z}}/S_{\mathbb{Z}}$ through its action on $\mathbb{P}^{n+v}$.

1.9. Proposition. *Let $g : Y \rightarrow T$ be a proper smooth morphism whose geometric fibers are complete intersections $V_n(\underline{a})$.*

- (i) *Étale-locally on T , Y is isomorphic to the intersection of v hypersurfaces $H_1 \dots H_v$, of degrees $a_1 \dots a_v$, in projective space $\mathbb{P}_T^{n+v}$ over T .*
- (ii) *In particular, étale-locally on T , there exist $u : T \rightarrow S_{\mathbb{Z}}$ and a T -isomorphism $\iota : Y \xrightarrow{\sim} u^*(X_{\mathbb{Z}}/S_{\mathbb{Z}})$.*
- (iii) *If (u', ι') is another system as in (ii), then, étale-locally on T , there exists $a \in G(T)$ such that $(u', \iota') = a(u, \iota)$.*
- (iv) *$S_{\mathbb{Z}}$ is smooth over $\mathrm{Spec}(\mathbb{Z})$.*

In this paragraph, the only corollary of 1.9 that we shall not use is the smoothness of $S_{\mathbb{C}}$. One could even dispense with it by subsequently replacing $S_{\mathbb{Z}}$ by the scheme $S'_{\mathbb{Z}}$ introduced below.

For every geometric fiber $Y_{\bar{t}}$ of g , one has $H^1(Y_{\bar{t}}, \mathcal{O}) = H^2(Y_{\bar{t}}, \mathcal{O}) = 0$, so that $\mathrm{Pic}_T(Y)$ is étale over T . Moreover, $\mathrm{Pic}(Y_{\bar{t}}) \simeq \mathbb{Z}$, generated by the class of the unique (very) ample invertible sheaf $\mathcal{O}(1)$ of degree $\prod a_i$. Étale-locally on T , there is therefore on Y a relatively ample invertible sheaf of degree $\prod a_i$, denoted $\mathcal{O}(1)$. Locally on T , it is unique. Since $H^1(Y_{\bar{t}}, \mathcal{O}(k)) = 0$ (FAC), $g_*\mathcal{O}(k)$ is locally free and its formation is compatible with base change ([9] p. 19 or EGA III); hence $\mathcal{O}(1)$ is very ample, and any choice of a basis of $g_*\mathcal{O}(1)$ gives a diagram

$$\begin{array}{ccc} Y & \xrightarrow{j} & \mathbb{P}_T^{n+v} \\ & \searrow g & \downarrow p \\ & & T \end{array}$$

with $j^*\mathcal{O}(1) = \mathcal{O}(1)$.

We prove that $j(Y)$ is an intersection of type (i); (iii) follows from the uniqueness of $\mathcal{O}(1)$. The maps $H^0(\mathbb{P}_{\bar{t}}^{n+v}, \mathcal{O}(k)) \rightarrow H^0(Y_{\bar{t}}, \mathcal{O}(k))$ are surjective, hence so are $p_*\mathcal{O}(k) \rightarrow g_*\mathcal{O}(k)$, and the kernel of this epimorphism of locally free sheaves is locally free; the assertion follows readily.

Let $\overline{S}'_{\mathbb{Z}}$ be the scheme over $\mathbb{Z}$ parametrizing v -tuples of hypersurfaces $H_i \subset \mathbb{P}^{n+v}$, with $\deg(H_i) = a_i$. Let $S'_{\mathbb{Z}}$ be the open subscheme of $\overline{S}'_{\mathbb{Z}}$ parametrizing the v -tuples (H_i) for which $\bigcap H_i$ is smooth of dimension n . The scheme $S'_{\mathbb{Z}}$ is smooth over $\text{Spec}(\mathbb{Z})$, being open in a product of projective spaces. The evident map from $S'_{\mathbb{Z}}$ to $S_{\mathbb{Z}}$ is surjective by (i). It is smooth (using (i), one readily verifies an infinitesimal criterion for smoothness). Hence $S_{\mathbb{Z}}$ is smooth over $\text{Spec}(\mathbb{Z})$.

Put $S = S_{\mathbb{C}}$, $X = X_{\mathbb{C}}$, and let $f : X \rightarrow S$ be the universal family 1.8.

1.10. Fundamental lemma. *Let $a : A \rightarrow S$ be an abelian scheme over S .*

(i) *If there exists a prime number ℓ and an isomorphism of $\mathbb{Q}_{\ell}$ -sheaves*

$$v_{\ell} : R^1 u_* \mathbb{Q}_{\ell} \xrightarrow{\sim} R^1 a_* \mathbb{Q}_{\ell},$$

or if there exists a horizontal isomorphism

$$v_{\text{DR}} : H^1_{\text{DR}}(J(X/S)/S) \xrightarrow{\sim} H^1_{\text{DR}}(A/S),$$

then A is isomorphic to $J(X/S)$. The isomorphism

$$v' : J(X/S) \longrightarrow A$$

is unique up to sign.

(ii) *If, in addition, A is endowed with a polarization in the principal series, then v' is compatible with the polarizations. If v_{ℓ} (respectively v_{DR}) as in (i) is compatible with the polarization forms, then v_{ℓ} (respectively v_{DR}) is induced by v' or by $-v'$.*

We shall use the following theorem of Lefschetz.

1.11. Theorem (Lefschetz). *Let $s \in S$, and let $X_s = f^{-1}(s)$.*

- (i) *The monodromy representation of $\pi_1(S, s)$ on $H^n(X_s, \mathbb{Q})$ is absolutely irreducible.*
- (ii) *For every prime number ℓ , the monodromy representation of $\pi_1(S, s)$ on $H^n(X_s, \mathbb{Z}/\ell)$ is absolutely irreducible.*

For an exposition in ℓ -adic cohomology of this theorem, following Lefschetz's proof very closely (cf. [6] Ch. V, III p. 106), see SGA 7 bis.

We prove 1.10. Consider the monodromy representations

$$\rho_J : \pi_1(S, s) \longrightarrow \text{GL}(H^1(J(X_s), \mathbb{Z})) \simeq \text{GL}(H^n(X_s, \mathbb{Z}))$$

$$\rho_A : \pi_1(S, s) \longrightarrow \text{GL}(H^1(A_s, \mathbb{Z})).$$

Hypothesis (i) implies that, after extension of scalars to $\mathbb{Q}_{\ell}$ or to $\mathbb{C}$, these representations become isomorphic. The isomorphism of two

representations is invariant under extension of the base field; hence there exists a $\pi_1(S, s)$ -equivariant isomorphism

$$v'' : H^1(J(X_s), \mathbb{Q}) \xrightarrow{\sim} H^1(A_s, \mathbb{Q}).$$

By 1.11(i), the image of $v''^{-1}(H^1(A_s, \mathbb{Z})) \cap H^1(J(X_s), \mathbb{Z})$ in $H^1(J(X_s), \mathbb{Z}/\ell)$ is all or nothing. It follows that a rational multiple of v'' is an isomorphism

$$v'_\mathbb{Z} : H^1(J(X_s), \mathbb{Z}) \xrightarrow{\sim} H^1(A_s, \mathbb{Z}).$$

By 1.11(i), $v'_\mathbb{Z}$ is uniquely determined up to sign.

By [3] 4.4.11 and 4.4.12, $v'_\mathbb{Z}$ comes from an isomorphism of abelian schemes

$$v' : J(X/S) \longrightarrow A,$$

which, like $v'_\mathbb{Z}$, is unique up to sign. We recall the principle of the proof. Interpret $v'_\mathbb{Z}$ as a section of the local system $\text{Hom}(R^1 u_* \mathbb{Z}, R^1 a_* \mathbb{Z})$; by 1.11(i), it is the unique section of this local system up to a scalar. It is also known ([3] 4.12) that the module of all sections defines, at each point of s , a Hodge substructure of $\text{Hom}(H^1(J(X_s), \mathbb{Z}), H^1(A_s, \mathbb{Z}))$. It follows that $v'_\mathbb{Z}$ is at each point of S of Hodge type $(0, 0)$, i.e. a morphism of Hodge structures, and the assertion follows.

We prove (ii). By 1.11(i), every polarization ψ' of $J(X/S)$ is a rational multiple $b {}_s\psi$ of the given polarization. If ψ' is in the principal series, b is a positive invertible integer, i.e. $\psi' = {}_s\psi$, which proves the first assertion. As for the second, a priori by 1.11(i) $v_\ell = b v'_\ell$ with $b \in \mathbb{Q}_\ell^*$ (respectively $v_{\text{DR}} = b v'_{\text{DR}}$ with $b \in \mathbb{C}^*$); compatibility with the forms ψ_ℓ or ψ_{DR} gives $b^2 = 1$, i.e. $b = \pm 1$.

1.12. By 1.11(i), the isomorphism α_ℓ is characterized up to sign by the fact that it carries ψ_ℓ to ${}_s\psi_\ell$ (1.6). It would be easy to adapt (and simplify) the arguments of [4] §6 to deduce from this that complete intersections $V_n(\underline{a})$ over finite fields satisfy the Weil conjectures. The same method would prove the following, perhaps more general, result.

1.13. Theorem. *Let*

$$\begin{array}{ccc} X & \xhookrightarrow{\quad} & \mathbb{P}_V^r \\ & \searrow & \downarrow \\ & & \text{Spec}(V) \end{array}$$

be a smooth projective variety of dimension $2m + 2$ over a discrete valuation ring V of unequal characteristic with finite residue field $\mathbb{F}_q$. Let X_0

and X_K be the special and general fibers of X . Suppose that the “new” part of the cohomology of the smooth hyperplane sections H of X_K has level ≤ 1 in the following sense.

(*) For $p + q = 2m + 1$ and $|q - p| > 1$, the cokernel

$$\text{Coker}(H^q(X_K, \Omega_{X_K}^p) \longrightarrow H^q(H, \Omega_H^p))$$

is zero.

Then, for every smooth hyperplane section H_0 of X_0 , the eigenvalues of the Frobenius endomorphism on

$$\text{Coker}(H^{2m+1}(X_0, \mathbb{Z}_\ell) \longrightarrow H^{2m+1}(H_0, \mathbb{Z}_\ell))$$

are algebraic integers all of whose complex conjugates have absolute value $q^{(2m+1)/2}$. These algebraic integers are divisible by q^m .

2. Over $\mathbb{Q}$

2.1. Let ℓ_0 be a prime number. For every extension K of $\mathbb{Q}$, consider the following problem:

(J, K) Construct a principally polarized abelian scheme $u : J_K \longrightarrow S_K$ over S_K (1.8), endowed with an isomorphism

$$\alpha_{\ell_0} : R^n f_* \mathbb{Z}_{\ell_0}(m) \xrightarrow{\sim} R^1 u_* \mathbb{Z}_{\ell_0}$$

which transforms ψ_{ℓ_0} , defined as in (1.6.2), into the polarization form.

For $K = \mathbb{C}$, problem (J, K) has, by 1.10, a solution unique up to unique isomorphism.

2.2. A general principle says that a problem P , defined over a subfield k of $\mathbb{C}$, and having over $\mathbb{C}$ a solution unique up to unique isomorphism, has a solution (equally unique) over k . Here this gives an abelian scheme $u : J_{\mathbb{Q}} \longrightarrow S_{\mathbb{Q}}$ over $S_{\mathbb{Q}}$, endowed with a principal polarization ${}_s\psi$ and an isomorphism α_{ℓ_0} as in (J, K). Moreover, the triple $(J_{\mathbb{Q}}, {}_s\psi, \alpha_{\ell_0})$ is unique up to unique isomorphism.

2.3. The preceding principle is justified as follows.

- If the problem P is to find a number a , with a having some geometric property, the principle is evident: k is the field of invariants of $\text{Aut}(\mathbb{C}/k)$, and a solution a of P over $\mathbb{C}$, being unique, is invariant under $\text{Aut}(\mathbb{C}/k)$. “Geometric” means: “invariant under extension of scalars.”
- If the problem P is to find a vector subspace of a vector space defined over k , having some geometric property, apply the same argument, in the form Bourbaki, Algebra, chap. 2, §8, Th. 1, to $\text{Aut}(\mathbb{C}/k)$. It is not required that V be finite-dimensional.

- c) If the problem P is to find a subscheme Y of an affine scheme W defined over k , having some geometric properties, apply b): the ideal of Y is a vector subspace of the coordinate ring.
- d) If, in c), W were not affine, one reduces to c) by gluing.
- e) For every integer $b \geq 3$, let M_b be the scheme over $S_{\mathbb{Z}[1/b]}$ whose sections over any $S_{\mathbb{Z}[1/b]}$ -scheme $t : T \rightarrow S_{\mathbb{Z}[1/b]}$ classify principally polarized abelian schemes $a : A \rightarrow T$ endowed with an isomorphism

$$t^* R^n f_* \mathbb{Z}/b(m) \xrightarrow{\sim} R^1 a_* \mathbb{Z}/b.$$

(This isomorphism is compatible with the intersection forms.)

A solution of problem (J, K) is interpreted as a section over S_K of

$$M_{\ell_0^\infty} = \varprojlim_c M_{\ell_0^c}.$$

Apply d) to this section, viewed as a subscheme of $(M_{\ell_0^\infty})_K$.

2.4. Remark. When problem P has a solution over a finite extension of k , the principle stated above is proved by Galois descent. One can often reduce to this case using the finite-extension principle EGA IV 9.1.1.

Let ℓ be a prime number. Let K be the function field of $S_{\mathbb{Q}}$, and let $\bar{K}$ be an algebraic closure of K . Every $\mathbb{Z}_\ell$ -sheaf F on $S_{\mathbb{Q}}$ defines an ℓ -adic representation of $\text{Gal}(\bar{K}/K)$ on the geometric generic fiber $F_{\bar{K}}$ of F .

2.5. Lemma. *The ℓ -adic representations of $\text{Gal}(\bar{K}/K)$ defined by $R^n f_* \mathbb{Z}_\ell(m)$ and by $R^1 u_* \mathbb{Z}_\ell$ have the same character.*

Let U be a Zariski open subset of $S_{\mathbb{Z}}$ on which ℓ and ℓ_0 are invertible, containing $S_{\mathbb{Q}}$, and such that $J_{\mathbb{Q}}$ is the restriction to $S_{\mathbb{Q}}$ of an abelian scheme $u' : J' \rightarrow U$. The ℓ -adic representations 2.5 of $\text{Gal}(\bar{K}/K)$ are then unramified outside U : they factor through a representation of $\pi_1(U, \bar{K})$. The same applies to their ℓ_0 -adic analogues. Each closed point s of U defines a conjugacy class of Frobenius substitutions $\varphi_s \in \pi_1(U, \bar{K})$. One has

$$\text{Tr}(\varphi_s^{-1}, (R^1 u_* \mathbb{Z}_\ell)_{\bar{K}}) = \text{Tr}(\varphi_s^{-1}, (R^1 u_* \mathbb{Z}_{\ell_0})_{\bar{K}}) \in \mathbb{Z} : \quad (2.5.1)$$

this is the trace of the Frobenius endomorphism of J'_s . Similarly,

$$\text{Tr}(\varphi_s^{-1}, (R^n f_* \mathbb{Z}_\ell(m))_{\bar{K}}) = \text{Tr}(\varphi_s^{-1}, (R^n f_* \mathbb{Z}_{\ell_0}(m))_{\bar{K}}) \in \mathbb{Q} : \quad (2.5.2)$$

and both sides equal, by the Lefschetz trace formula in ℓ -adic cohomology,

$$\frac{1}{q^m} (\#\mathbb{P}(k(s)) - \#X_s(k(s))).$$

By hypothesis,

$$\mathrm{Tr}(\varphi_s^{-1}, (R^1 u_* \mathbb{Z}_{\ell_0})_{\overline{K}}) = \mathrm{Tr}(\varphi_s^{-1}, (R^n f_* \mathbb{Z}_{\ell_0}(m))_{\overline{K}}),$$

whence, by (2.5.1) and (2.5.2),

$$\mathrm{Tr}(\varphi_s^{-1}, (R^1 u_* \mathbb{Z}_{\ell})_{\overline{K}}) = \mathrm{Tr}(\varphi_s^{-1}, (R^n f_* \mathbb{Z}_{\ell}(m))_{\overline{K}}).$$

One concludes by observing that the trace is a continuous function on $\pi_1(U, \overline{K})$, and that, by Čebotarev's density theorem, the Frobenius elements φ_s are dense in $\pi_1(U, \overline{K})$.

2.6. Lemma. *The representations of $\mathrm{Gal}(\overline{K}/K)$ on $R^n f_* \mathbb{Q}_{\ell}(m)$ and on $R^n f_* \mathbb{Z}/\ell(m)$ are irreducible.*

Indeed, the image of $\mathrm{Gal}(\overline{K}/K)$ in $\mathrm{GL}(R^n f_* \mathbb{Q}_{\ell}(m))$, or in $\mathrm{GL}(R^n f_* \mathbb{Z}_{\ell}(m))$, contains the image of the geometric fundamental group $\pi_1(S_{\overline{\mathbb{Q}}}, \overline{K})$. By 1.11, the action of the latter is already irreducible.

From 2.5 and 2.6 it follows that the ℓ -adic representations of $\mathrm{Gal}(\overline{K}/K)$ on $(R^n f_* \mathbb{Z}_{\ell}(m))_{\overline{K}}$ and on $(R^1 u_* \mathbb{Z}_{\ell})_{\overline{K}}$ are isomorphic. Every isomorphism between these representations comes from an isomorphism

$$\alpha'_{\ell} : R^n f_* \mathbb{Z}_{\ell}(m) \xrightarrow{\sim} R^1 u_* \mathbb{Z}_{\ell}.$$

By 1.11, after extending scalars from $\mathbb{Q}$ to $\mathbb{C}$, α'_{ℓ} becomes an ℓ -adic multiple of the isomorphism (1.6.1): $\alpha'_{\ell} = b\alpha_{\ell}$, with $b \in \mathbb{Z}_{\ell}^*$. Over $\mathbb{Q}$, put $\alpha_{\ell} = b^{-1}\alpha'_{\ell}$.

2.7. We have thus obtained:

- a) a principally polarized abelian scheme $J_{\mathbb{Q}}$ over $S_{\mathbb{Q}}$;
- b) for every prime number ℓ , an isomorphism compatible with the intersection forms

$$\alpha_{\ell} : R^n f_* \mathbb{Z}_{\ell}(m) \xrightarrow{\sim} R^1 u_* \mathbb{Z}_{\ell};$$

- c) an isomorphism

$$J_{\mathbb{Q}} \times_{S_{\mathbb{Q}}} S_{\mathbb{C}} \xrightarrow{\sim} J(X_{\mathbb{C}}/S_{\mathbb{C}})$$

which identifies the preceding isomorphisms with the isomorphisms (1.6.1).

2.8. For every extension K of $\mathbb{Q}$, let W_K be the K -vector space of horizontal morphisms

$$\alpha' : H_{\mathrm{DR}}^n(X_K/S_K) \longrightarrow H_{\mathrm{DR}}^1(J_K/S_K).$$

One has $W_K = W_{\mathbb{Q}} \otimes K$. For $K = \mathbb{C}$, $W_{\mathbb{C}}$ is one-dimensional, and its nonzero elements are isomorphisms. Hence there exists an isomorphism

$$\alpha' : H_{\mathrm{DR}}^n(X_{\mathbb{Q}}/S_{\mathbb{Q}}) \xrightarrow{\sim} H_{\mathrm{DR}}^1(J_{\mathbb{Q}}/S_{\mathbb{Q}}).$$

Moreover, α' is unique up to multiplication by $\lambda \in \mathbb{Q}^*$. Over $\mathbb{C}$, one verifies that

- a) α' respects the Hodge filtrations on the two sides: $\alpha'(F^{m+1}) = F^1$;
- b) α' transforms ψ_{DR} into a multiple of ${}_s\psi_{\text{DR}}$:

$$\alpha'(\psi_{\text{DR}}) = \mu(\alpha') {}_s\psi_{\text{DR}} \quad (\mu(\alpha') \in \mathbb{Q}^*).$$

One has $\mu(\lambda\alpha') = \lambda^{-2}\mu(\alpha')$.

2.9. Proposition. *The multiplier $\mu(\alpha')$ is positive.*

This is checked after extension of scalars from $\mathbb{Q}$ to $\mathbb{R}$.

2.10. Conjecture. *The multiplier $\mu(\alpha')$ is a square.*

This conjecture is equivalent to the following statements.

$$\text{For a suitable choice of } \alpha', \text{ one has } \alpha'(\psi_{\text{DR}}) = {}_s\psi_{\text{DR}}. \quad (2.10.1)$$

$$\text{Over } \mathbb{C}, \text{ the isomorphism deduced from } \alpha' \text{ transforms rational cohomology into rational cohomology.} \quad (2.10.2)$$

$$\text{Via 2.7c), the isomorphism } \alpha_{\text{DR}} \text{ (1.7.1) is defined over } \mathbb{Q}. \quad (2.10.3)$$

That this conjecture follows from both the Hodge conjecture and the Tate conjecture is meagre consolation.

Recall that

$$G = \text{PGL}_{n+v+1} \quad \text{acts on} \quad X_{\mathbb{Q}}/S_{\mathbb{Q}} \quad (1.8).$$

2.11. Lemma. *There exists one and only one action of G on $J_{\mathbb{Q}}/S_{\mathbb{Q}}$ extending the action of G on $S_{\mathbb{Q}}$. For this action, the isomorphisms α_{ℓ} (2.7.b)) are equivariant.*

For every extension K of $\mathbb{Q}$, consider the following problem.

(I, K) . Let $\mu : G_K \times_K S_K \longrightarrow S_K$ be the action of G_K on S_K . Find an isomorphism

$$a : \mu^* J_K \xrightarrow{\sim} \text{pr}_2^* J_K$$

which defines an action of G_K on J_K/S_K . Since G is connected, the α_{ℓ} are automatically equivariant. From this, or from Mumford [9] 6.1 p. 115, it follows that a , if it exists, is unique. By the principle stated in 2.2, it is enough to show that a exists over $\mathbb{C}$. This follows from the fact that, over $\mathbb{C}$, the intermediate Jacobian is defined up to unique isomorphism and from the fact that $\mu^*(X_K/S_K)$ is given as isomorphic to $\text{pr}_2^*(X_K/S_K)$.

From 1.9 and 2.11 one obtains the following theorem.

2.12. Theorem. *Up to unique isomorphism, there exists one and only one functor J of the following type, endowed with the following additional data.*

- a) *For T a scheme of characteristic 0 and $g : Y \longrightarrow T$ a proper smooth morphism with fibers $V_n(\underline{a})$, $u : J(Y/T) \longrightarrow T$ is a principally polarized abelian scheme.*
- b) *J is compatible with base changes $c : T' \longrightarrow T$: isomorphisms $J(c^*Y/T') \xrightarrow{\sim} c^*J(Y/T)$ are given.*

c) *Compatible base-change isomorphisms*

$$\alpha_\ell : R^n g_* \mathbb{Z}_\ell(m) \xrightarrow{\sim} R^1 u_* \mathbb{Z}_\ell$$

are given.

d) *When restricted to smooth schemes over $\mathbb{C}$, the functor $(J, (\alpha_\ell))$ becomes isomorphic to the “intermediate Jacobian” functor constructed in §1.*

3. Over $\mathbb{Z}$

The theory over $\mathbb{Z}$ is as unsatisfactory as it is in De Rham cohomology (2.8 to 2.10).

3.1. Proposition. *There exists a proper birational morphism $b : \tilde{S} \rightarrow S_{\mathbb{Z}}$ such that*

- a) *b induces an isomorphism of $\tilde{S}_{\mathbb{Q}}$ with $S_{\mathbb{Q}}$;*
- b) *there exists a principally polarized abelian scheme $\tilde{J}$ over $\tilde{S}$, whose restriction to $S_{\mathbb{Q}}$ is $J_{\mathbb{Q}}$.*

We use the following lemma.

3.2. Lemma. *Let V be a discrete valuation ring with fraction field K of characteristic 0, and let $g : \text{Spec}(V) \rightarrow S_{\mathbb{Z}}$ be a morphism. Then the abelian variety over K obtained by pullback from $J_{\mathbb{Q}}$ has good reduction.*

Let $\bar{K}$ be an algebraic closure of K , let X_V be the inverse image of X over V , let J_K be the inverse image of J over $\text{Spec}(K)$, and let $X_{\bar{K}}$ and $J_{\bar{K}}$ be the inverse images of X_V and J_K over $\bar{K}$.

Let $k \geq 3$ be an integer invertible on V . Since X_V is smooth over V , the specialization theorem in ℓ -adic cohomology implies that the $\text{Gal}(\bar{K}/K)$ -module $H^n(X_{\bar{K}}, \mathbb{Z}/k)(m)$ is unramified. By 2.7, this $\text{Gal}(\bar{K}/K)$ -module is isomorphic to $H^1(J_{\bar{K}}, \mathbb{Z}/k)$, and it remains to apply the Néron–Ogg–Shafarevich criterion for good reduction.

A short, but too complicated, method for deducing 3.1 from 3.2 is the following. Let $k \geq 3$, and let M_k be as in 2.3.e). Then $J_{\mathbb{Q}}$ defines a section of M_k over $S_{\mathbb{Q}}$. Let $\tilde{S}_k$ be the closure of this section in M_k . To prove that $b : \tilde{S}_k \rightarrow S_{\mathbb{Z}[1/k]}$ is proper, it is enough to test the valuative criterion of properness on discrete valuation rings whose fraction fields have characteristic 0; indeed, $\tilde{S}_{k\mathbb{Q}}$ is dense in $\tilde{S}_k$. For these rings, it suffices to apply 3.2.

It is easy to check that, over $\text{Spec}(\mathbb{Z}[1/k_1 k_2])$, $\tilde{S}_{k_1}$ coincides with $\tilde{S}_{k_2}$. The $\tilde{S}_k$ glue to form the desired $\tilde{S}$.

3.3. I conjecture that in 3.1 one may take $\tilde{S} = S$. If this were so, 2.12 would remain true over $\mathbb{Z}$.

One verifies easily:

3.4. Proposition. *Let $a : \tilde{J} \longrightarrow \tilde{S}$ be as in 3.1. For each prime number ℓ , the isomorphism α_ℓ (2.7) extends over $\tilde{S}[1/\ell]$ to*

$$\alpha_\ell : R^n f_* \mathbb{Z}_\ell(m) \xrightarrow{\sim} R^1 a_* \mathbb{Z}_\ell.$$

3.5. The Weil conjecture for the $V_n(\underline{a})$.

Using 3.4 one can give a variant of the proof, indicated in 1.12, of the Weil conjecture for the $V_n(\underline{a})$.

Let X be a smooth complete intersection of dimension n and multidegree $\underline{a}$ in $\mathbb{P}^{n+v}(\mathbb{F}_q)$. X corresponds to a point $x \in S_{\mathbb{Z}}(\mathbb{F}_q)$.

a) *there exists a point $\tilde{x} \in \tilde{S}(\mathbb{F}_q)$ above x .*

Indeed, one may lift X to a complete intersection over the Witt vectors $W(\mathbb{F}_q)$, which lifts x to $x' \in S_{\mathbb{Z}}(W(\mathbb{F}_q))$. The point x' defines a point of $\tilde{S}(W(\mathbb{F}_q) \otimes \mathbb{Q})$, and, since b is proper, this point extends to $x'' \in \tilde{S}(W(\mathbb{F}_q))$. Reducing x'' gives $\tilde{x}$.

b) Let J be the abelian variety over $\mathbb{F}_q$, the fiber of $\tilde{J}$ at $\tilde{x}$. If $\bar{X}$ and $\bar{J}$ are obtained from X and J by extension of scalars from $\mathbb{F}_q$ to one of its algebraic closures $\bar{\mathbb{F}}_q$, then 3.4 gives an isomorphism

$$H^n(\bar{X}, \mathbb{Z}_\ell)(m) \xrightarrow{\sim} H^1(\bar{J}, \mathbb{Z}_\ell).$$

This isomorphism commutes with the (geometric) Frobenius endomorphisms, so

$$\det(1 - Ft, H^n(\bar{X}, \mathbb{Z}_\ell)) = \det(1 - q^m Ft, H^1(\bar{J}, \mathbb{Z}_\ell)). \quad (3.5.1)$$

The Weil conjecture for X therefore follows from a theorem of Weil [11] for J .

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